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[23CSE301]

THIRD SEMESTER B.Tech EXAMINATION, DECEMBER-2024

OPERATING SYSTEMS

Time: 3 Hours **Maximum Marks: 100**

Instructions: i. Missing data may be suitably assumed.

ANSWER ALL QUESTIONS

		PART-A	**5 x2=10**		
1.	**a.**	Define Operating Systems and its functions.	L1	CO1	**02**
	b.	Draw the process life cycle model.	L2	CO2	**02**
	c.	Define Thrashing.	L1	CO3	**02**
	d.	What are the types of Access Methods?	L2	CO4	**02**
	e.	Distinguish between IoS and Android.	L1	CO6	**02**
		PART-B	**6 X 5=30**		
2.	**a.**	Explain the concept of Multiprocessor and Multicoreorganization.	L2	CO1	**05**
		OR			
	b.	Discuss the functionality of system boot with respect to anoperating system.	L2	CO1	**05**
3.	**a.**	State critical section problem. Discuss three solutions to solve the critical section problem.	L2	CO2	**05**
		OR			
	b.	Distinguish among short-term, medium-term and long-term scheduling with suitable example.	L2	CO2	**05**
4.	**a.**	Elaborate about the free space management on I/O buffering and blocking.	L2	CO3	**05**
		OR			
	b.	Given memory partitions of 100KB, 500KB, 200KB, 300KB and 600KB (in order), how would each of the first-fit, best-fit and worst-fit algorithms place processes of 212KB, 417KB,112KB and 426KB (in order)? Which algorithm makes the most efficient?	L3	CO3	**05**
5.	**a.**	Explain directory structure.	L2	CO4	**05**
		OR			
	b.	Explain file system mounting in detail.	L2	CO4	**05**
6.	**a.**	Explain in detail the memory management in LINUXoperating system.	L2	CO5	**05**

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		OR			
	b.	Discuss the requirements needed to become a Linux System Administrative.	L2	CO5	**05**
7.	**a.**	Draw the architecture of IoS.	L2	CO6	**05**
		OR			
	b.	Describe in detail the file system.	L2	CO6	**05**
		PART-C			**6 X 10=60**
8.	**a.**	Explain different operating system structures with neatsketch.	L2	CO1	**10**
		OR			
	b.	Explain the various types of system calls with examples.	L2	CO1	**10**
9.	**a.**	Explain the FCFS, preemptive and non-preemptive versions of Shortest-Job First scheduling algorithms with Gantt charts for the four Processes given. Compare their average turnaround and waiting time.	L3	CO2	**10**

Process	ArrivalTime	WaitingTime
P1	0	8
P2	1	4
P3	2	9
P4	3	5

	OR											
b.	Consider the following values of a system: P0-P4 are 5 processes present and A, B, C, D are the resources. The maximum need of a process and the allocated resources details are given in the table. Answer the following based on banker's algorithm											

	Allocation				Max				Available			
	A	B	C	D	A	B	C	D	A	B	C	D
P_0	0	0	1	2	0	0	1	2	1	5	2	0
P_1	1	0	0	0	1	7	5	0				
P_2	1	3	5	4	2	3	5	6				
P_3	0	6	3	2	0	6	5	2				
P_4	0	0	1	4	0	6	5	6				

i. What is the content of NEED matrix?
ii. Is the system in a safe state?
iii. Which processes may cause deadlock if the system is not safe.
iv. If a request from process p1 arrives for (0, 4, 3,1)can the request be granted immediately? Justify.

L3 CO2 **10**

10.	**a.**	Consider page reference string 6, 1, 1, 2, 0, 3, 4, 6, 0, 2, 1, 2, 1, 2, 0, 3, 2, 1, 2, 0 with 3 pages frames. Find the number of page faults for the algorithms FIFO and LRU.	L3	CO3	**10**
		OR			
	b.	Explain the concept of demand paging in detail with neatdiagram.	L2	CO3	**10**
11.	**a.**	Consider the following disk request sequence for a disk with 100 tracks 45, 21, 67, 90, 4, 50, 89, 52, 61, 87, 25. Head pointer starting at 50 and moving in left direction. Find the number of head movements in cylinders using FCFS scheduling, SSTF Scheduling.	L3	CO4	**10**
		OR			
	b.	Discuss the Linked List and Indexed File Allocation methods with neat diagram. Mention their advantages anddisadvantages.	L2	CO4	**10**
12.	**a.**	Explain Inter-Process Communication (IPC) in Linux.	L2	CO5	**10**
		OR			
	b.	Explain in detail the design principles, kernel modules,process management, scheduling in LINUX system.	L2	CO5	**10**
13.	**a.**	Describe an android operating system with architecture.	L2	CO6	**10**
		OR			
	b.	Compare the feature of IoS and android.	L2	CO6	**10**